DESIGN SPEC · PURCHASING · FEATURE II · GEO-SCOPED SUPPLIER SEARCH

Small orders stay local. Big orders *travel farther*.

Nexi filters supplier suggestions by distance from a configurable base, and the search radius scales up with the estimated order value. The bigger the ₱, the wider she looks — because a large order justifies the transport a small one doesn't.

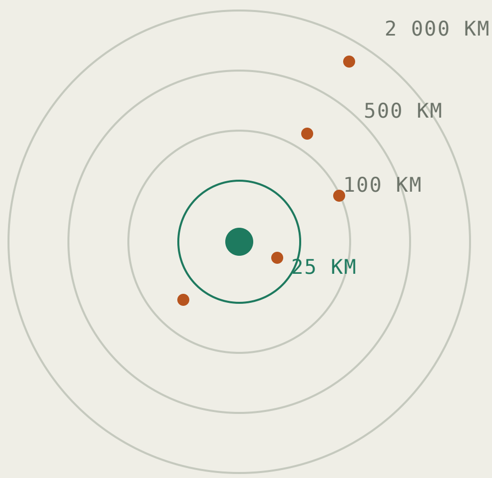
Module · Accounting → Purchasing
Feature · geo-scoped RFQ search
Branch · claude/accounting-purchasing-suppliers-5p94sz
Head · 4fac878
Doc rev · 2026-07-27

§ 1

The rule, in one sentence

Value determines geography.

A ₱4,000 pack of consumables shouldn’t be sourced from another country. A ₱1.2M capex order shouldn’t be limited to the barangay. The system encodes that judgement so Nexi doesn’t need to be reminded, and every buyer follows the same rule.



The rings are the tiers. The base pin is your farm. Each ring is the radius unlocked by a higher order-value bucket. Nexi searches inside the smallest ring the order qualifies for, and widens only if it can’t find enough candidates.

Rings are illustrative, not to scale — a real 2 000 km ring wouldn’t fit next to a 25 km ring on the same page.

§ 2

Default tier ladder

Sensible starting values — every one editable.

UP TO ₱	RADIUS	MEANING · WHEN IT FIRES
10,000	**25 km**	**Small · local only.** Consumables, replacements, urgent top-ups. Freight would dwarf the order.
50,000	**100 km**	**Regular · metro.** Weekly resupply, mid-size components. Same-day / next-day trucking still economic.
200,000	**500 km**	**Mid · regional.** Bulk media, quarterly packaging runs. Willing to wait a few days for better price.
1,000,000	**2,000 km**	**Large · national.** Equipment, project-scale inputs. Whole-country search opens meaningful competition.
∞	**∞**	**Very large · worldwide.** Capex, imports, one-off large-ticket. International freight is a rounding error against the deal.

Ladder is edited in the settings modal: `📍 Location & radius`. Rows can be added, removed, relabelled, and re-priced; the top row is always the ∞ catch-all. Ordering is by *maxValue* — the first row whose ceiling is ≥ the order value wins.

§ 3

How the value is inferred

The buyer types intent, not arithmetic.

The outreach modal exposes an `Est. value ₱` field. It's not required — Nexi guesses:

- › **From the SOS hint** · if the product matches a SOS item with a `purchaseCost`, estimated value = `qty × lastCost`. This is the primary path once SOS is connected.
- › **From past history** · if any past PO line has the same product name, the median past unit cost × current qty is used as a fallback (roadmap).
- › **Buyer override** · typing a number in the field wins over any guess. The Override km field wins over the tier the value picks. Rules serve the buyer, not the other way around.

§ 4

Auto-widen safety

Nexi never returns zero suppliers because of the geography rule.

Every tier has a floor: `minSuppliersForRadius` (default **3**). If the requested radius surfaces fewer than that many contactable suppliers, Nexi automatically promotes the search to the next larger tier and re-runs.

> BEHAVIOUR ON WIDEN
>
> The status strip in the modal flags the widen explicitly — *"auto-widened · fewer than min in the requested tier"* — so the buyer sees when the rule bent and can adjust the tier or the min-count without hunting.

The floor is per-tier by design; the config lives on the base-location object so it moves with the site.

§ 5 Distance computation

Great-circle, not driving distance.

`_purchDistanceKm(lat1, lng1, lat2, lng2)` uses the Haversine formula. It’s the right choice for a geography filter: cheap, deterministic, no external service, correct within a fraction of a percent at country-scale distances.

It is *not* driving distance. For a “how far is the truck actually going” number, a future step could call a routing API — but for a filter that decides whether to invite a supplier to quote, great-circle is more than accurate enough, and one straight-line rule is easier to explain than a road network that changes.

§ 6

Base location

One pin. Editable. Optional browser geolocation.

Set once from the header button `📍 Location & radius`. The modal captures a display name, an address, latitude and longitude — and offers a `📡 USE MY CURRENT LOCATION` button that reads `navigator.geolocation` and fills the coords in place.

Default until edited: Manila metro at `14.5995, 120.9842`. That's a sensible pin for AteretPaz; other sites edit it once and forget it.

```
// hydroPro_acct_purch_base_location
{
  name: "AteretPaz farm",
  address: "",
  lat: 14.5995,
  lng: 120.9842,
  tiers: [
    { maxValue: 10000,   radiusKm: 25,   label: "Small · local or
    { maxValue: 50000,   radiusKm: 100,  label: "Regular · metro'
    { maxValue: 200000,  radiusKm: 500,  label: "Mid · regional"
    { maxValue: 1000000, radiusKm: 2000, label: "Large · national
    { maxValue: Infinity, radiusKm: null, label: "Very large · wo
  ],
  minSuppliersForRadius: 3,
  includeUnknown: true
}
```

§ 7

Supplier coordinates

Per-supplier lat / lng, with graceful fallback.

A dedicated key stores explicit coordinates per supplier — keyed by lowercased name so a rename in one place doesn't sever the link.

```
// hydroPro_acct_purch_supplier_locations
{
  "micropoint graphics": {
    lat:  14.6532,
    lng:  121.0498,
    address: "Quezon City",
    city:    "Quezon City",
    country: "PH"
  },
  …
}
```

Fallback path. If a supplier has no explicit coord record, `_purchGetSupplierLocation()` reaches into the existing `hydroPro_inv_suppliers_v1` and `hydroPro_acct_vendors` stores and returns any address / city / country found there — so the row can still show something even if the distance chip has to say *"no location · add"*.

Adding coords in-flow. Every row in the outreach modal shows a 📍 chip. Rows without coordinates get a clickable `no location · add` chip that prompts for lat / lng inline; the answer is written straight into the map and the modal re-renders. No mode-switching to a settings page.

ROADMAP

Coord entry is manual today because the app can't geocode from the browser (no server-side routing service). A future `/nexi/geocode?address=` route on the hnx-sync worker would let Nexi auto-fill coords from any address on the vendor record.

§ 8

The outreach modal, changed

Two new controls, one new column, one status strip.

A

Est. value ₱

Auto-guessed from qty × SOS last-cost. Editable. Feeds the tier lookup.

B

Override km

Any positive number wins over the tier. Blank means “use the tier”.

C

Status strip

🎯 Radius X km · tier Y · in-range vs outside count · widen flag · base name + coords.

D

📍 chip / row

Every supplier row shows distance. In-range rows are full opacity, outside rows are dimmed.

E

Selectors

Top 3 · Existing · All-in-range — every bulk selector respects the in-range flag.

Nothing is hidden by the filter — out-of-range rows are still visible, just dimmed and unchecked by the bulk selectors, so the buyer can override on a specific supplier without editing the tier ladder.

§ 9

Nexi can be asked directly

Same rule, natural-language surface.

ASK NEXI	WHAT SHE RETURNS
“suppliers near me”	Ranked list by distance from the base, top 5, with contact info and past-order counts.
“local suppliers for rockwool”	Same, filtered to the product word. Falls back to closest few if no one is inside the default radius.
“suppliers within 50 km”	Parses the number, honours the override, returns everyone inside that radius.
“where is my base”	Reads the base-location record — name and coords.

New keywords: `radius`, `nearby`, `near me`, `local`, `distance`, `km`, `kilometer`, `area`. They’re appended to the domain’s keyword array so the registry routes those questions to the geo answerer before falling through to the price / supplier answerers.

§ 10

Storage summary

Two keys added, everything else re-used.

KEY	PURPOSE
`hydroPro_acct_purch_base_location`	Base name + coords + tier ladder + auto-widen floor + unknown-inclusion flag.
`hydroPro_acct_purch_supplier_locations`	Map of supplier-lower → {lat, lng, address, city, country}.
`hydroPro_inv_suppliers_v1`	Read only for fallback address / city / country when a supplier has no explicit coords.
`hydroPro_acct_vendors`	Same fallback role as above.

`Infinity` can't survive a JSON round-trip, so on save it's written as `null` in the top tier's `maxValue`; on load the config-loader restores it. Any consumer that reads the config through `_purchGetBase()` never sees the sentinel.

§ 11

Not doing yet

Extension points, called out.

DEFERRED	WHAT UNLOCKS IT
Geocode a plain address	`/nexi/geocode?address=` on the hnx-sync worker. Would auto-fill supplier coords from stored addresses on Vendors master.
Driving-distance / actual travel time	OSRM or Google Distance Matrix through the worker. Only worth adding if lead-time promises depend on it — otherwise great-circle is enough.
Freight-cost aware radius	Multiply `qty × unitCost` by a freight-per-km rate before the tier lookup, so the tier reflects *delivered* cost, not just *ordered* cost.
Per-category tier overrides	Attach a `categoryOverrides` map on the base — e.g. "perishables never go past 200 km, regardless of value".
Multi-site (per-farm) bases	Promote the base record to an array keyed by site; the outreach modal picks the site the buyer selects.

None of the above breaks the current model. Each extension either adds a stored field, adds a worker route, or plugs a new signal into the same tier lookup.

HydroNexis-AI · Accounting · Purchasing · Geo-scoped supplier search

Design revision 2026-07-27 · commit `4fac878`